

Lida Xu

lidaxu66@umd.edu | +1 (240) 564-1841 | [Website](#)
ORCID: 0009-0005-3655-7868 | [Google Scholar](#) | [LinkedIn](#) | [GitHub](#)

Updated: September 1, 2026

Research Summary

My research asks whether the mathematical structure that makes quantum materials immune to disorder can be turned into an engineering resource in photonics — frequency combs that tolerate fabrication defects, harmonic generation that needs no poling, and optical computing architectures that scale. I work at the intersection of topological physics, nonlinear optics, and integrated photonics, combining theory, simulation, and chip-scale experiment to turn topological protection from an elegant idea into a working tool.

Education

- 2021.8–2026.12 **Ph.D. in Physics** (expected), University of Maryland, College Park, USA — Dissertation: *Nonlinear topological photonics with coupled microresonators*. Advisor: Mohammad Hafezi; co-advisors: Kartik Srinivasan (NIST), Yanne K. Chembo.
- 2020.6–2021.8 **Research Assistant** (gap year), Southern University of Science and Technology, China — Advisor: Xiuhao Deng.
- 2016.9–2020.6 **B.Sc. in Physics**, Nanjing University, China — GPA 91/100. Thesis: *Characterization of a single-photon source for rare-earth-ion-doped quantum memory*. Advisor: Xiaosong Ma.

Publications & Preprints († equal contribution / co-first author)

Published

- Science** 390 (6773), 612–616 (2025). *Multi-timescale frequency-phase matching for high-yield nonlinear photonics*. M. J. Mehrabad†, **L. Xu**†, G. Moille, C. J. Flower, S. Sarkar, A. Padhye, S. C. Ou, D. G. S. Forero, M. Ghafarizadeh, Y. K. Chembo, K. Srinivasan, M. Hafezi.
Press: [Phys.org](#), [Optics.org](#), [SciTechDaily](#), [MIT Technology Review \(China\)](#)
Contribution: led the experimental demonstration of nested frequency-phase matching.
- Science Advances** 11, eadw7696 (2025). *On-chip multi-timescale spatiotemporal optical synchronization*. **L. Xu**†, M. J. Mehrabad†, C. J. Flower†, G. Moille, A. Restelli, D. G. S. Forero, Y. Chembo, S. Mittal, K. Srinivasan, M. Hafezi.
Contribution: led the experimental demonstration and theoretical analysis of multi-timescale mode locking.
- Science Advances** 11, eadv2023 (2025). *Sub-wavelength optical lattice in 2D materials*. S. Sarkar, M. J. Mehrabad, D. G. S. Forero, L. Gu, C. J. Flower, **L. Xu**, K. Watanabe, T. Taniguchi, S. Park, H. Jang, Y. Zhou, M. Hafezi.
Press: [JQI News](#)
Contribution: designed the grating couplers via 3D-FDTD simulation.
- Science** 384 (6702), 1356–1361 (2024). *Observation of topological frequency combs*. C. J. Flower†, M. J. Mehrabad†, **L. Xu**†, G. Moille, D. G. S. Forero, O. Örsel, G. Bahl, Y. Chembo, K. Srinivasan, S. Mittal, M. Hafezi.
Press: [Phys.org](#)
Contribution: led the theoretical simulation and the large-scale device search (217 devices).

Submitted / under review

- Single-shot realization of 10,000-mode octave-spanning artificial gauge fields*. **L. Xu**†*, A. Padhye†, S. Sarkar, A. Parhizkar, C. J. Flower, G. Moille, K. Srinivasan, M. Hafezi, M. J. Mehrabad.
(*corresponding author) — [arXiv:2606.23960](#)
Contribution: conceived the core idea, built the theoretical model, and led the experimental demonstration.
- Quantum metamorphosis: programmable emergence and the breakdown of bulk–edge dichotomy in multiscale systems*. M. J. Mehrabad†, A. Parhizkar†, **L. Xu**†, G. Moille, A. Dutt, D. Englund, K. Srinivasan, D. Leykam, M. Hafezi.
[arXiv:2511.13831](#)
Contribution: built the software framework for quantum metamorphosis and carried out the nonlinear simulations.
- Stability analysis of nested solitons in topological microresonator lattices*. S. Akter, **L. Xu**, M. J. Mehrabad, M. Hafezi, C. R. Menyuk.
Contribution: provided the core code and used perturbation theory to predict the loss channels of the supermodes.

Patents & Invention Disclosures

1. *TOPAI: Topological photonic architectures for optical computing and artificial intelligence.*
M. J. Mehrabad, **L. Xu**, S. Sarkar, Z. Y. Wei, M. Hafezi. U.S. provisional patent 63/964,690 (filed January 2026).
Winner, University of Maryland Invention of the Year 2026 (Quantum Technology category).
2. *Nested frequency and phase matching.*
M. J. Mehrabad, **L. Xu**, G. Moille, K. Srinivasan, M. Hafezi. U.S. provisional patent 63/944,606 (filed December 2025).
3. *Quantum optical metamorphosis.*
M. J. Mehrabad, A. Parhizkar, **L. Xu**, M. Hafezi. Invention disclosure, UM Ventures (August 2025).
4. *Systems of ultrabroadband multimodal artificial gauge fields.*
M. J. Mehrabad, **L. Xu**, S. Sarkar, A. Padhye, M. Hafezi. Invention disclosure, UM Ventures (August 2025).

Research Funding

1. **MPower Early Scholars Investment Fund**, University of Maryland 2026
University-wide competitive bridge funding (approx. 20 projects funded), up to \$30,000, to advance dissertation milestones and pursue sustained external funding.
2. **NSF I-Corps™ Teams**, Spring 2026 National Cohort (NY I-Corps Hub) 2026
Entrepreneurial Lead. Award: \$50,000. PI: M. Hafezi. NSF award OMA-2120757 (“Robust Quantum Simulation”).
Commercialization of topological coupled-microresonator arrays for nonlinear photonic integrated circuits in sensing and photonic computing. Conducted 100+ customer-discovery interviews during the seven-week national cohort.

Conference Presentations & Talks

Invited

1. Physics of Quantum Electronics (PQE) Winter Colloquium, 2026.
2. Joint Quantum Institute (JQI) Seminar, University of Maryland, 2025.

Contributed

1. *Integrated nonlinear topological photonics.* **L. Xu**, A. Padhye, S. Sarkar, A. Parhizkar, C. J. Flower, G. Moille, *et al.* — APS Division of Atomic, Molecular and Optical Physics (DAMOP), 57th Annual Meeting, 2026.
2. *Frequency comb realization of the integer quantum Hall model.* **L. Xu**, A. Padhye, S. Sarkar, A. Parhizkar, G. Moille, *et al.* — CLEO: Fundamental Science, FM2B.3, 2026.
3. *Wafer-scale integrated nonlinear optics based on nested coupled ring-resonator lattices.* A. Padhye, M. J. Mehrabad, **L. Xu**, S. Sarkar, A. Parhizkar, C. J. Flower, S. C. Ou, *et al.* — CLEO: Fundamental Science, FM2D.6, 2026.
4. *Nested frequency-phase matching for high-yield nonlinear photonics.* M. J. Mehrabad, **L. Xu**, G. Moille, C. J. Flower, S. Sarkar, S. C. Ou, *et al.* — CLEO: Fundamental Science, FM2G.6, 2026.
5. *Diagnosing chaos in topological super-ring Kerr microcombs via Lyapunov exponents.* S. Akter, **L. Xu**, M. J. Mehrabad, M. Hafezi, C. R. Menyuk — CLEO: Fundamental Science, FW1B.5, 2026.
6. *Quantum metamorphosis: topology in multiscale systems.* A. Parhizkar, M. J. Mehrabad, **L. Xu**, M. Hafezi, A. Dutt, G. Moille, D. Leykam — APS Global Physics Summit, 2026.
7. *Topological frequency combs.* C. J. Flower, M. J. Mehrabad, **L. Xu**, G. Moille, K. Srinivasan, S. Mittal, M. Hafezi — CLEO: Science and Innovations, SF3J.1, 2024.

Awards & Honors

1. **Invention of the Year**, University of Maryland — winner, Quantum Technology category 2026
2. **Ralph Myers and Friends of Physics Award**, honorable mention 2021
3. **Elite (Top-Notch) Program Scholarship**, School of Physics, Nanjing University 2018–2019

Teaching & Mentoring

1. **Teaching Assistant**, Experimental Physics I (Mechanics and Thermodynamics), University of Maryland — prepared laboratory apparatus and guided undergraduates through the experiments.
2. **Research mentoring**, JQI / University of Maryland — mentored several junior graduate students on theoretical and experimental projects in topological photonics and integrated nonlinear photonics.

Professional Service & Outreach

1. **Peer reviewer** — *Nature Photonics*, *Science Advances*, *Nature Communications*, *Physical Review Letters*.
2. **Science communication** — independently produced a physics video series on Bilibili; the [Dirac series](#) has over 60,000 cumulative views.

Technical Skills

Photonic design & simulation: Lumerical FDTD, Tidy3D, transfer-matrix (TMM) and FDFD solvers, 3D-FDTD grating couplers, particle-swarm inverse design, microresonators and coupled-ring arrays, dispersion engineering

Layout & fabrication interface: KLayout, GDSFactory; PDK-compliant, DRC-clean device cells for Si_3N_4 tape-outs; design-to-measurement iteration on returned chips

Experiment: fiber-chip coupling, tunable-laser spectroscopy, OSA/ESA/oscilloscope characterization, electro-optic and RF drive electronics, automated test and data acquisition

Theory & modeling: Lugiato-Lefever equation, split-step Fourier methods, coupled-mode theory, Floquet/Bogoliubov stability analysis, $\chi^{(2)}/\chi^{(3)}$ phase matching, frequency combs and soliton dynamics, quantum Hall physics

Programming: Python (NumPy, SciPy, JAX, PyQt5, GPU acceleration, concurrency), MATLAB, Git **Typesetting:** $\text{\LaTeX}$ **Graphics:** Inkscape

Languages: English (fluent), Mandarin Chinese (native)

References

Prof. Mohammad Hafezi — Ph.D. advisor
hafezi@umd.edu

Prof. Kartik Srinivasan — Ph.D. co-advisor
kartik.srinivasan@nist.gov

Prof. Yanne K. Chembo — Ph.D. co-advisor
ykchembo@umd.edu